

COMPUTER SCIENCE TRIPOS Part IB, Part II 50% – 2022 – Paper 7

5 Formal Models of Language (apc38)

This question concerns the noisy channel framework for problem-solving in natural language processing.

(a) You have the following sentences translated from English into Triposi.

English	Triposi
she drinks water	mwamni sileng
the rain soaks the teacher	sileng mworob sesesrakan
the teacher drinks here	sesesrakan mwamni mwabma
the teacher keeps drinking	sesesrakan mwatbo mwamni

(i) Translate the following English sentences into Triposi:

the rain keeps soaking the teacher

she keeps drinking water here

[2 marks]

translation

Answer: 1 mark for each correct translation (n.b. these are not real translations and so cannot be searched online: Triposi is based on a real language (Abma), as are the words featured in this exercise, but the translations are made up) –
 sileng mwatbo mworob sesesrakan
 mwatbo mwamni sileng mwabma

(ii) Describe how we can calculate the likelihood of a translation using the noisy channel framework: you will need to give and explain the equation for decoding from one language into another, and explain how you can obtain the information needed to carry out the calculations. [4 marks]

noisy channel
translation

Answer: 1 mark for relating the equation for noisy channel decoding of a translated language: $\arg\max p(f)p(e|f)$ (or by presenting f as I for ‘input’ and e as O for ‘output’, where the aim is to decode a de-noised version of the source sentence I'). 1 mark for presenting $p(f)$ as a model of ‘fluency’, or the ‘language model’. 1 mark for presenting $p(e|f)$ as a model of ‘fidelity’, or the ‘translation model’. 1 mark for referring to frequencies from training corpora.

(iii) What problem do the following underlined English words present, given the training data we have so far, and how can you still translate them into Triposi with a machine?

the teachers drink Perrier

[3 marks]

translation of
unseen words

Answer: 1 mark for describing the problem as ‘unknown’, ‘unseen’ or ‘out-of-vocabulary’ words. 1 mark for either proposing that ‘teachers’ is translated using ‘teacher’, or for proposing that ‘teachers’ could be translated with access to morphological information – we know how to translate ‘teacher’ and so need a rule to derive its plural form in Triposi. 1 mark for either proposing a baseline approach for ‘Perrier’ where it is copied into the target language as a naive translation method, or for referring to a statistical contingency plan such as ‘smoothing’ or related techniques used in NLP (but not themselves lectured in this course – therefore the intuition for

handling unseen events and preventing zero probabilities would be enough, without the technical detail). Also vector semantic approaches are possible, whereby translation could be possible via ‘water’ since it is a known word and might be the most similar to ‘Perrier’ if we have a large corpus of these words used in context. Or, observing that ‘sileng’ means both ‘water’ and ‘rain’ and therefore could be used to represent ‘Perrier’ too, is fine and would earn the 1 mark.

- (b) You know that your optical character recognition model has made errors on every predicted instance of ‘e’. Explain what information you need in order to automatically correct the errors with a noisy channel approach. [3 marks]

noisy channel
error correction

Answer: 1 mark for mentioning n-gram frequencies, 1 mark for confusion counts for the character ‘e’ (will also accept approaches involving edit distance to known words). 1 mark for reference to a training corpus for the retrieval of this information (referring to a dictionary for an edit distance approach is also acceptable).

- (c) A signaller sends you Morse code messages, but you know that they send a dot when they should send a dash two times in five, and a dash instead of a dot three times in ten. You also know that they use the character M (– –) 3 times in 100, N (– ·) 7/100, and A (· –) 8/100.

You receive the message “· · – · · –”. What is the likelihood that it represents MIN, MAN, NAN or AIM? [4 marks]

noisy channel
processing

Answer: The correct calculations are given below. 1 mark for indicating that character frequencies of the hypothesised word should be multiplied together. 1 mark for indicating that dash/dot error rates should be multiplied together. 1 mark for showing all four likelihood calculations correctly. 1 mark for having the correct primary candidate word: AIM.

word	code	likelihood
MIN	– – · · – ·	$.03 * .07 * .07 * .4 * .4 * .3 * .6 * .4 * .3 = 5.08032e-07$
MAN	– – · – – ·	$.03 * .08 * .07 * .4 * .4 * .3 * .4 * .4 * .3 = 3.87072e-07$
NAN	– · · – – ·	$.07 * .08 * .07 * .4 * .6 * .3 * .4 * .4 * .3 = 1.354752e-06$
AIM	· – · · – –	$.08 * .07 * .03 * .6 * .4 * .3 * .6 * .4 * .7 = 2.032128e-06$

- (d) Discuss the similarities and differences between the noisy channel and human processing of spoken English. [4 marks]

language
processing

Answer: Open-ended. 1 mark for each relevant point. **Similarities** include: there can literally be a noisy channel in human conversation (e.g. background noise, a bad phone-line, a cocktail party), prior linguistic expectations (a ‘language model’) are used to understand what is said, speech decoding makes use of both acoustic and linguistic information, speech processing makes use of historical information (what has been said so far / n-gram modelling). **Differences** include: the use of other information sources to understand the speech signal (visual cues from the other speaker such as gesture and facial expression, objects and actions in the local surroundings, world knowledge, conversation context and interpersonal relationships), speakers might make accidental errors and not correct them, speakers might make errors and partially repeat preceding sequences to correct them, speakers might be interrupted and not finish what they were saying, listeners might not be paying full attention. Note that reference

simply to ‘use of context’ as a difference is insufficient unless it’s more precisely stated that this would be longer-term conversational context (because short-term context is modelled through n-grams).
